

Characterizing the Survival-Associated Interactions Between Tumor-Infiltrating Lymphocytes and Tumors From Pathological Images and Multi-Omics Data

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Abstract—The tumor-infiltrating lymphocytes (TILs) and its correlation with tumors have shown significant values in the development of cancers. Many observations indicated that the combination of the whole-slide pathological images (WSIs) and genomic data can better characterize the immunological mechanisms of TILs. However, the existing image-genomic studies evaluated the TILs by the combination of pathological image and single-type of omics data (e.g., mRNA), which is difficulty in assessing the underlying molecular processes of TILs holistically. Additionally, it is still very challenging to characterize the intersections between TILs and tumor regions in WSIs and the high dimensional genomic data also brings difficulty for the integrative analysis with WSIs. Based on the above considerations, we proposed an end-to-end deep learning framework *i.e.*, IMO-TILs that can integrate pathological image with multi-omics data (*i.e.*, mRNA and miRNA) to

analyze TILs and explore the survival-associated interactions between TILs and tumors. Specifically, we firstly apply the graph attention network to describe the spatial interactions between TILs and tumor regions in WSIs. As to genomic data, the Concrete AutoEncoder (*i.e.*, CAE) is adopted to select survival-associated Eigengenes from the high-dimensional multi-omics data. Finally, the deep generalized canonical correlation analysis (DGCCA) accompanied with the attention layer is implemented to fuse the image and multi-omics data for prognosis prediction of human cancers. The experimental results on three cancer cohorts derived from the Cancer Genome Atlas (TCGA) indicated that our method can both achieve higher prognosis results and identify consistent imaging and multi-omics biomarkers correlated strongly with the prognosis of human cancers.

Index Terms—Tumor-infiltrating lymphocytes (TILs), multi-omics, survival analysis, Graph Attention Networks (GATs).

I. INTRODUCTION

WITH the development of cancer biology, scientists have discovered that the progression of cancers are not only affected by the unlimited growth of tumor cells but also supported and nurtured in complex and dynamic tumor micro-environment (*i.e.*, TME) [1]. TME is consisted of various types of cells, including fibroblasts and a wide spectrum of immune cells that influence cancer cell behavior [2]. Among different types of TME components, the tumor-infiltrating lymphocytes (*i.e.*, TILs) reflect the host anti-cancer immune response and is verified to have significant associations with the clinical outcome of cancer patients [3]. For example, recent studies [4] suggested the tumor-infiltrating B cells and T cells act as an important marker for prognosis in different cancer subtypes. Fridman et al. [5] reported that the high TILs density in colorectal cancer usually enabled favorable clinical outcomes.

Early studies on the analysis of the immune mechanism of TILs are mainly based on whole-slide pathological images (*i.e.*, WSI) or omics data. For example, Saltz et al. [6] have presented the mappings of TILs based on whole-slide pathological images (WSIs) from 13 cancer cohorts, and validated that the spatial characterizations of TILs are associated with the

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diagnosis and prognosis of human cancers. Leon-Ferre et al. [7] discussed the effects of immune cells on tumor progression, and identified that the distributions of TILs are associated with the overall survival of breast cancer. Moreover, Xu et al. [8] have proposed a novel deep learning algorithm to quantify the TIL densities in pathological images and Lea et al. [9] have built a computational framework to determine the locations of TILs in pathological images, which can help identify subgroups of colorectal carcinoma patients with distinct survival.

On the other hand, the rapid advancement in high-throughput sequencing technologies has enabled the collection of various types of “omics” data with unprecedented speed [10], and many computational techniques have been applied to analyze TILs and its correlation with tumors from omics profiles. For instance, Zhou et al. [11] proposed a machine learning-based computational framework for the recognition of the miRNA signatures of TILs. Liu et al. [12] analyzed the TIL patterns of 993 melanoma samples and screened differential expression genes related to diverse immune subtypes. Other studies include [13] used the hierarchical clustering algorithm to explore the TIL-related gene signatures and finally identified seven survival-associated genomic features.

In many clinical trials, it is common to acquire both imaging and omics data that can offer both spatial or molecular descriptions for cancer patients [14], [15]. Recently, several studies also endeavored to integrate WSIs with omics data to analyze the roles of TILs for the prognosis of human cancers. For example, Lu et al. [16] firstly generated the cell-level TIL maps with 43 quantitative image features, and then combined them with the mRNA data to perform survival analysis on breast cancer. Binder et al. [17] have presented a computational pipeline for the integrated profiling of whole-slide pathological images, omics and clinical features to analyze TILs and its correlation with tumors. These studies suggested that different TIL bio-markers (*i.e.*, image and genomic data) provide complementary information, which is useful for the prognosis task when used together.

Although much progress have been achieved, the existing studies generally have the following three limitations. Firstly, most of the existing image-genomic association studies evaluated the TILs by the combination of pathological image and single type of omics data (*i.e.*, mRNA data). However, the mRNA data can only reflect the gene regulation mechanism at transcriptional and translational levels [18], [19], which cannot assess the underlying molecular processes of TILs holistically. As a matter of fact, the existing studies demonstrated that the miRNA-mediated post-transcriptional gene regulation can also affect the interactions between TILs and tumor cells [20]. Intuitively, the joint investigation of pathological image data with multi-omics data (*i.e.*, mRNA and miRNA) can better characterize the association between TILs and tumors, and identify more comprehensive bio-markers for cancer prognosis. Secondly, it is still very challenging to characterize the intersections between TILs and tumors in WSIs because of their large size and heterogeneity patterns, and most existing efforts focused on designing statistical measurements [16] or unsupervised machine learning algorithms [6] to describe such spatial interactions at WSI level, which is difficulty in analyzing the survival-associated interactions between TILs

and tumors in view of micro-scale *i.e.*, patch-level. Thirdly, most of the integrative studies simply combined the spatial and molecular profiles of TILs for the following survival analysis task [16], [21], which neglected to take the intrinsic correlation among WSI and multi-omics data into consideration. As a matter of fact, the exploration of the associations among multi-modal data has been widely accepted as a key component of current multi-modality based machine learning approaches [22].

Based on the above considerations, in this paper, we proposed an end-to-end deep learning framework *i.e.*, IMO-TILs that can integrate pathological image with multi-omics data (*i.e.*, mRNA and miRNA) to analyze TILs and explore the survival-associated interactions between TILs and tumors. Specifically, for WSIs data, we use the graph attention network (*i.e.*, GAT) to describe the patch-level spatial interactions of TILs and tumor across WSIs. As to each type of multi-omics data, we apply co-expression network analysis algorithms to cluster genes into co-expressed modules followed by applying the Concrete AutoEncoder to select survival-associated modules. Finally, we introduce the attention mechanism to the deep generalized canonical correlation analysis (DGCCA) algorithm that can simultaneously capture the correlations among WSIs and multi-omics data and fuse the multi-modal data for the final clinical outcome prediction task. We implement the proposed IMO-TILs algorithm on three cancer cohorts derived from The Cancer Genome Atlas (TCGA), the experimental results indicate that our method can not only achieve better prognosis results than the comparing methods, but also identify consistent survival-associated imaging and multi-omics bio-markers correlated strongly with the patients’ survival.

Our preliminary work combining WSIs with single-type of omics data (*i.e.*, mRNA data) to characterize the immunological mechanisms of TILs was published on MICCAI 2022 [21]. In this substantially expanded journal paper, we offer new contributions in the following aspects: 1) Besides mRNA data, the extended model also incorporated miRNA data, which can better stratify cancer patients with distinct survival than our preliminary work. 2) Extending the model by applying the deep generalized canonical correlation analysis (DGCCA) algorithm to explore the associations between image and multi-omics data. 3) Evaluating the effectiveness of the proposed method on two additional datasets (*i.e.*, Kidney renal clear cell carcinoma (KIRC) and Lung Squamous Cell Carcinoma (LUSC) datasets). 4) In-depth explanations of the imaging and multi-omics bio-markers that are identified by the proposed IMO-TILs model.

II. MATERIALS AND METHODS

A. Overview of Our Method

Fig. 1 shows the flowchart of our method, which are generally consisted of four steps. Firstly, we pre-processed the collected pathological images and multi-omics data (*i.e.*, mRNA and miRNA). Then, the graph attention network is applied to characterize the interactions between TILs and tumor patches. Next, for each type of omics data, our method adopts the Concrete AutoEncoder to identify survival-associated gene modules. Finally, the deep generalized canonical correlation analysis (DGCCA) accompanied with the attention layer is

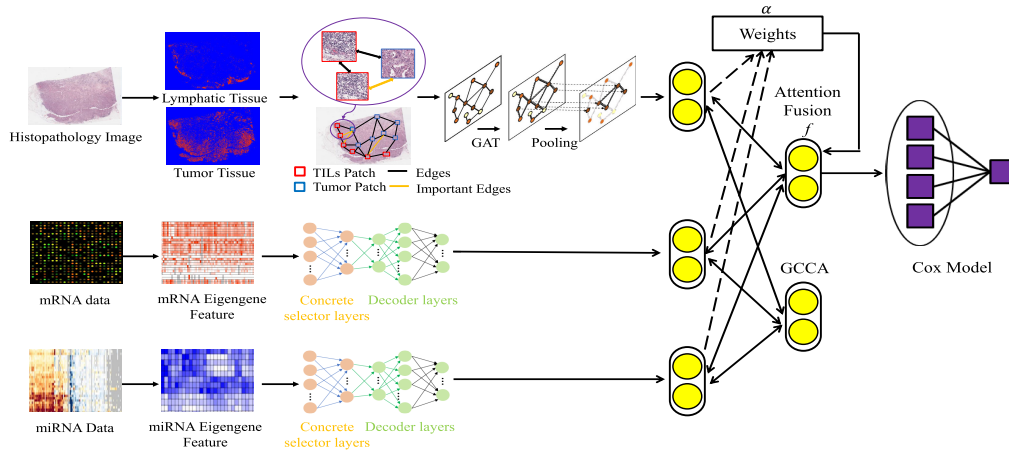


Fig. 1. The framework of our study including four parts, *i.e.*, the pre-processing of the WSIs and genomic data, the characterization of the interaction between TILs and tumor patches by GAT, the identification of important Eigengenes by Concrete AutoEncoder, and the integrative analysis of imaging and multi-omics features for human survival analysis.

TABLE I

DEMOGRAPHICS AND CLINICAL INFORMATION OF THE DATASET	BRCA	LUSC	KIRC
Number of Patients:	365	276	279
Censored Non-censored	333/32	170/106	131/148
Age (Mean)	57.6	68.04	58.80
Median Follow-up Time (Month)	27.3	32.8	34.3

implemented to fuse the image and multi-omics data for prognosis prediction of human cancers.

B. Dataset

The Cancer Genome Atlas (TCGA) project has generated paired histology and multi-omics data, as well as detailed clinical information for different types of cancers [23]. In this paper, we collect matched pathological imaging and multi-omics data (*i.e.*, miRNA and mRNA) on three cancer cohorts *i.e.*, Breast Invasive Carcinoma (BRCA), Kidney Renal Clear Cell Carcinoma (KIRC) and Lung squamous cell carcinoma (LUSC) derived from TCGA. Table I summarizes the demographic information of all the patients we use in this work, where censored patients mean that the death events of these patients are not observed during their follow-up, while the non-censored category corresponds to the patients whose recorded survival times are the exact time from initial diagnosis to death.

C. Data Pre-Processing

Pathological Image Pre-processing: Since each of the collected whole-slide pathological image is of large size (*e.g.*, 100,000-by-100,000 pixels). We first divide each WSI into 512×512 patches without overlap to facilitate processing. Then, by considering the generated patches may contain background regions that do not contain enough tissue, we select the patches whose image density are larger than 0.7 for further analysis. Here, the image density is calculated as the percentage of non-white (at least one of the red, green, and blue values was below 200 in the 24-bit RGB color space) pixels in that patch [24]. Next, we construct a U-net++ network based on the annotated dataset released by Amgad et al. [25] to segment the TIL and tumor tissues. Specifically, by considering the uneven distribution of different tissues, we applied

the Lovasz-Softmax loss for U-net++ network training [26]. After getting the pixel-level segmentation results of each valid patch, we calculate its tumor and TILs area ratios and select 200 patches with the largest tumor or TILs ratio, respectively.

Multi-omics Data Pre-Processing: For each type of multi-omics data (*i.e.*, mRNA and miRNA), instead of directly examining the high-dimensional omics data, we apply the local maximum quasi-clique merging (ImQCM) algorithms introduced in [27] to identify densely connected gene modules using singular value decomposition and summarize each module as an Eigengene. For mRNA data in each cancer cohort, we set ImQCM parameters $\gamma = 0.7, \lambda = 1, t = 1, \beta = 0.4, c = 10$, and adopt Spearman's rank correlation coefficient to calculate mRNA-wised correlations. Finally, we get 57, 56 and 50 Eigengenes for BRCA, KIRC and LUSC cohorts, respectively. As to miRNA data, the parameters of ImQCM are fixed as $\gamma = 0.7, \lambda = 1, t = 1, \beta = 0.6, c = 4$, which finally yield to 15, 17 and 17 miRNA eigengenes for BRCA, KIRC and LUSC cohorts, respectively.

D. Characterize the Interactions Between TILs and Tumor Patches by Graph Attention Network

Based on the extracted tumor and TILs patches in each WSI, we apply the Graph Attention Network (GAT) to describe the interactions among them. Specifically, given patches as vertices for each WSI, vertex features are generated by the pre-trained ResNet-101 model on ImageNet, followed by implementing principal component analysis (PCA) to reduce their dimensions to 64. Then, we adopt the K-Nearest Neighbors (KNN) algorithm to construct the graph basing on the Euclidean distances between patch pairs, and the number of nearest neighbors for each node is set as 20. The inputs for GAT network are a set of node features, *i.e.*, $h = \{h_1, h_2, \dots, h_N\}$, where $h_i \in \mathbb{R}^d$ and N represents the number of extracted patches in each WSI. The GAT layer produces a new set of node features $h' = \{h'_1, h'_2, \dots, h'_N\}$ via linear transformation matrix $W \in \mathbb{R}^{d \times d'}$ and $h'_i \in \mathbb{R}^{d'}$. Then, a shared attentional mechanism $\mathbb{R}^{d'} \times \mathbb{R}^{d'} \rightarrow \mathbb{R}$ is applied to calculate the attention coefficients between two node i and j as follows:

$$e_{ij} = a(W h_i, W h_j) \quad (1)$$

Moreover, GAT normalizes e_{ij} 's values of node j by applying the Softmax function:

$$\alpha_{ij} = \text{softmax}_j(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{u \in N_i} \exp(e_{iu})} \quad (2)$$

where N_i denotes the neighborhood node for node i . Finally, the output features for every node can be represented as:

$$h'_i = \sigma \left(\sum_{j \in N_i} \alpha_{ij} W h_j \right) \quad (3)$$

In addition, in order to avoid over-fitting and identify survival-associated interactions among different patches, we adopt SAGPool [28] to perform local pooling operations of node embeddings after each GAT layer. In SAGPool, the contribution of each node embedding in the pooling receptive field to the next network layer is learned using an attention mechanism, and the attention score $Z \in R^N$ in each graph can be calculated via:

$$Z = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H' \Theta_{att}) \quad (4)$$

where $H' = [h'_1, h'_2, \dots, h'_N]^T \in R^{N \times d'}$, σ is the activation function and $\tilde{A} \in R^{N \times N}$ is the adjacency matrix for each graph with self-connections, $\tilde{D} \in R^{N \times N}$ is the degree matrix of $\tilde{A}$, and $\Theta_{att} \in R^{d'}$ is the only parameter of the SAGPool layer. After getting the self-attention score Z for each node, SAGPool utilizes the method introduced in [29] to select top s ($s \in [0, 100]$) percentage of nodes with high attention scores. At the last layer of the GAT, the mean pooling strategy is applied to aggregate the node features into WSI-level feature.

E. Identification of Important Eigengenes by Concrete AutoEncoder

With the gene co-expression modules derived from ImQCM, we employ the Concrete AutoEncoder (CAE) to identify survival-associated features [30] for each type of omics data. Different from the traditional AutoEncoder, CAE also includes the Concrete selector layer (CSL) that can define the number of features (*i.e.*, k) to be selected, and the selected features are parameterized by $\alpha \in R^q$. Then, each element of the sample m from the Concrete distribution is defined as:

$$m_j = \frac{\exp((\log \alpha_j + g_j)/T)}{\sum_{k=1}^d \exp((\log \alpha_k + g_k)/T)} \quad (5)$$

where m_j refers to the j^{th} element for m . T is a temperature parameter and $g \in R^q$ is a q -dimensional vector sampled from a Gumbel distribution. As the temperature (*i.e.*, T) of the concrete selector layer (CSL) approximates to zero, the concrete distribution approaches to discrete distribution, the sampled weights will converge to one-hot vectors. The feature selection strategy in CSL is implemented in the following ways. For each of the k nodes in CSL, we sample a d -dimensional concrete random variable $Z^{(i)}$, $i \in \{1, 2, \dots, k\}$, and then the i -th node in the selector layer outputs $u(i) = XZ^{(i)}$ for the input data X . It is worth noting that $Z^{(i)}$ will approximate to one-hot vector if T approximates to 0, and thus we can totally obtain k features by the combination of different $u(i)$, $i = 1, 2, \dots, k$.

F. Deep Generalized Canonical Correlation Analysis (DGCCA)

After applying GAT and Concrete AutoEncoder to extract multi-view features from pathological images and multi-omics data, it is common to integrate them for better cancer assessment and risk prediction. However, most of the existing studies directly concatenate them into a long feature vector for the following survival analysis [31], [32], which neglects to take the intrinsic relationship across different views into consideration. As a matter of fact, the exploitation of multi-modal association has been widely accepted as a key component of current multi-modality based machine learning approaches [33], [34]. Based on the above considerations, we applied deep generalized canonical correlation analysis (DGCCA) framework [35], [36] that can capture the correlation among imaging and multi-omics data. Specifically, let $f_P(X_P)$, $f_M(X_M)$, $f_I(X_I)$ represent the output of the final layer for pathological image, mRNA and miRNA data, respectively. Here, $f_i(X_i) = [f_i(X_i^1), f_i(X_i^2), \dots, f_i(X_i^N)] \in R^{o_i \times N}$, $i \in \{P, M, I\}$, where N indicates the number of samples and o_i represents the dimension for each type of data. The goal of DGCCA is to solve the following optimization problem:

$$\begin{aligned} \min_{U_i, G} \quad & \sum_{i \in \{P, M, I\}} \|G - U_i^T f_i(X_i)\|_2^2 \\ \text{s.t.} \quad & GG^T = I \end{aligned} \quad (6)$$

where $U_i \in R^{o_i \times r}$ and $G \in R^{r \times N}$. The aim of DGCCA is to learn appropriate representation function for each view *i.e.*, $f_i(X_i)$, $i \in \{P, M, I\}$ for the purpose of approximating the projection of each view *i.e.*, $U_i^T f_i(X_i)$ to the shared representation G , and thus the inter-correlations across different views can be captured. Here, we provide the sketch for calculating derivative of the DGCCA loss with respect to $f_i(X_i)$, more details can be found in [35]. In particular, we define $C_{ii} = f_i(X_i) f_i(X_i)^T \in R^{o_i \times o_i}$ as the covariance matrix for the i -th view, and $P_i = f_i(X_i) C_{ii}^{-1} f_i(X_i)^T \in R^{N \times N}$ corresponds to the projection matrix applied for data normalization. Here, we define $M = \sum_{i \in \{P, M, I\}} P_i$, and it is easy to infer that M is positive semi-definite since each P_i is positive semi-definite. Also, it is obviously that G is consisted of top r eigenvectors of M , and $U_i = C_{ii}^{-1} f_i(X_i) G^T$. Accordingly, the objective function of DGCCA can be reformulated as:

$$\begin{aligned} & \sum_{i \in \{P, M, I\}} \|G - U_i^T f_i(X_i)\|_2^2 \\ &= \sum_{i \in \{P, M, I\}} \|G - G f_i(X_i)^T C_{ii}^{-1} f_i(X_i)\|_2^2 \\ &= 3r - \text{Tr}(GMG^T) \end{aligned} \quad (7)$$

Since $3r$ are constant term, the minimization of Eq. (7) can be transformed to minimize $L_{DGCCA} = -\text{Tr}(GMG^T)$, which equals to the sum of eigenvalues of M . By taking the derivative of L_{DGCCA} with respect to $f_i(X_i)$, we have

$$\frac{\partial L_{DGCCA}}{\partial f_i(X_i)} = -2U_i G + 2U_i U_i^T f_i(X_i) \quad (8)$$

As shown in Eq. (8), if the shared representation G is far away from the view-specific representation $U_i^T f_i(X_i)$, then its corresponding feature extraction network $f_i(X_i)$ should receive a large update and vice versa.

G. DGCCA With Attention Mechanism for Integrative Survival Analysis

Besides applying the DGCCA algorithm to capture the correlation among multi-view data. We further adopt an attention-mechanism-based feature fusion method [37] for integrative survival analysis of human cancers. Specifically, let $f_i(X_i^j) \in R^{o_i}$ be the output for i -th view on sample j . Then, the joint representations for sample j , i.e., $g(X^j)$ can be obtained by the weighted sum of $f_i(X_i^j)$, $i \in \{W, M, I\}$:

$$g(X^j) = \sum_{i \in \{P, M, I\}} a_i^j V_i f_i(X_i^j) \quad (9)$$

where

$$a_i^j = \frac{\exp\{w^T \tanh(V_i f_i(X_i^j))\}}{\sum_{i \in \{P, M, I\}} \exp\{w^T \tanh(V_i f_i(X_i^j))\}} \quad (10)$$

where $V_i \in R^{o \times o_i}$ and $w \in R^o$ are trainable parameters. The attention-based multi-view fusion strategy allows to assign different weights to different views, which is more flexible than the existing maximum or mean pooling strategy for multi-view data fusion.

After the attention-based multi-view fusion step, we generate $g(X^j) \in R^o$ for each sample. Here, we applied the widely used Cox proportional hazard model [38] to conduct the survival analysis of cancer patients, whose negative partial likelihood can be formulated as:

$$l_{Surv} = - \sum_{j=1}^N \delta_j (\beta^T g(X^j) - \log(\sum_{k \in R(t_j)} \exp(\beta^T g(X^k)))) \quad (11)$$

where $\delta_j = 1$ or $\delta_j = 0$ indicates a non-censored or a censored patient. $R(t_j)$ represents the set of patients that are still under risk before time t_j . According to Eqs. (9)-(10), we can optimize $f_i(X_i^j)$ by minimizing Eq. (11), and the derivative of L_{Surv} with respect to $f_i(X_i^j)$ can be represented as:

$$\frac{\partial L_{Surv}}{\partial f_i(X_i^j)} = \frac{\partial L_{Surv}}{\partial g(X^j)} \frac{\partial g(X^j)}{\partial f_i(X_i^j)} \quad (12)$$

It is easy to infer that

$$\frac{\partial L_{Surv}}{\partial g(X^j)} = -\beta + \frac{\beta \exp(\beta^T g(X^j))}{\sum_{k \in R(t_j)} \exp(\beta^T g(X^k))} \quad \text{if } \delta_j = 1 \quad (13)$$

Meanwhile,

$$\frac{\partial g(X^j)}{\partial f_i(X_i^j)} = a_i^j V_i^T + a_i^j (1 - a_i^j) V_i^T Q w (V_i f_i(X_i^j))^T \quad (14)$$

where $Q \in R^{o \times o}$ is a diagonal matrix with the m -th element $Q_{mm} = 1 - \tanh(T_m)^2$. Here, T_m equals to the m -th element

of $V_i f_i(X_i^j)$. Then, the derivative of L_{Surv} with respect to $f_i(X_i)$ is denoted as:

$$\frac{\partial L_{Surv}}{\partial f_i(X_i)} = [\frac{\partial L_{Surv}}{\partial f_i(X_i^1)}, \frac{\partial L_{Surv}}{\partial f_i(X_i^2)}, \dots, \frac{\partial L_{Surv}}{\partial f_i(X_i^N)}] \quad (15)$$

By the combination of Eq. (8) and Eq. (15), we finally get the total gradient of $f_i(X_i)$ for the loss function of the proposed IMO-TILs method as:

$$\frac{\partial L}{\partial f_i(X_i)} = \frac{\partial L_{Surv}}{\partial f_i(X_i)} + \lambda \frac{\partial L_{DGCCA}}{\partial f_i(X_i)} \quad (16)$$

where λ is the regularization parameter used for balancing different terms.

III. EXPERIMENTAL RESULTS

A. Experimental Settings

To evaluate the performance of the proposed IMO-TILs framework for the prognosis prediction of human cancers, we test it on three cancer cohorts (i.e., BRCA, LUSC and KIRC) derived from TCGA. For each cancer cohort, we apply the five fold cross-validation strategy to evaluate the performance of our method. In each fold of the cross-validate, we split the 20% samples from the training dataset as the validation set. For the GAT network used for characterizing the interactions between TILs and Tumor patches, we design a neural network consisted of 2 GAT layers and 2 graph pooling layers. The dimensions for the GAT layers are set as 512 and 256 respectively, and we select top 80% of nodes with the highest attention score for each graph pooling layer. For each type of omics data, we apply Concrete AutoEncoder (CAE) to select important Eigengenes with one concrete selector layer (CSL). Here, the number of neurons in CSL is set as $k = 10$. Finally, the regularization parameter (shown in Eq. (16)) applied to balance survival and DGCCA losses is fixed as $1e-3$. We evaluate the performance of our proposed IMO-TILs method by the measurements of Concordance Index (i.e., CI) and AUC [40]. The CI and AUC values range from 0 to 1, and the larger values of CI and AUC indicate better prognosis performance, and vice versa.

B. Integrative Survival Analysis of Human Cancers Based on the Combination of Pathological Images and Multi-Omics Data

We first test the performance of our proposed IMO-TILs model based on the combination of pathological images, mRNA and miRNA data. Table II shows the prognostic results of our method, compared with the following approaches only using the spatial information of TILs, single type of omics data or the combination of multi-omics data for survival analysis of human cancers. 1) Lu et al. [16](Im): Extract 43 quantitative image features from global cell-level TIL maps for survival analysis of human cancers. 2) Saltz et al. [6](Im): An automatic pipeline that can extract tissue-level TILs features from the pathological images for clinical outcome prediction. 3) DeepSurv(mR) [42] or DeepSurv(miR): deep learning generalization of the Cox proportional hazards model applying on mRNA or miRNA data. 4) I-TILs(Im): A variant of the proposed IMO-TILs, which only using the TILs spatial information and their intersections with tumors for survival analysis

TABLE II

INTEGRATIVE SURVIVAL ANALYSIS OF HUMAN CANCERS BASED ON THE COMBINATION OF TILs SPATIAL INFORMATION AND MULTI-OMICS DATA. HERE, 'Im', 'mR' AND 'miR' REFER TO THE IMAGE, mRNA AND miRNA DATA, RESPECTIVELY. THE "*" IN EACH ROW INDICATES THAT THE PROPOSED IMO-TILs IS SIGNIFICANTLY SUPERIOR ($p < 0.05$) TO THE COMPETING METHOD VIA T-TEST

	BRCA		LUSC		KIRC		P-value	
	CI	AUC	CI	AUC	CI	AUC	CI	AUC
Lu et al [16](Im)	0.601 ± 0.047	0.621 ± 0.031	0.611 ± 0.021	0.637 ± 0.022	0.621 ± 0.052	0.641 ± 0.037	*	*
Saltz et al(Im) [6]	0.581 ± 0.041	0.597 ± 0.037	0.601 ± 0.052	0.627 ± 0.041	0.618 ± 0.041	0.634 ± 0.027	*	*
DeepSurv(mR) [39]	0.577 ± 0.039	0.611 ± 0.041	0.561 ± 0.047	0.591 ± 0.021	0.589 ± 0.051	0.619 ± 0.041	*	*
DeepSurv(miR) [39]	0.586 ± 0.046	0.601 ± 0.036	0.556 ± 0.039	0.564 ± 0.028	0.581 ± 0.027	0.611 ± 0.039	*	*
I-TILs(Im)	0.644 ± 0.031	0.665 ± 0.052	0.639 ± 0.034	0.671 ± 0.024	0.643 ± 0.029	0.678 ± 0.018	*	*
O-TILs(mR)	0.629 ± 0.056	0.651 ± 0.053	0.625 ± 0.033	0.643 ± 0.056	0.623 ± 0.058	0.656 ± 0.064	*	*
O-TILs(miR)	0.616 ± 0.058	0.634 ± 0.055	0.601 ± 0.032	0.638 ± 0.049	0.605 ± 0.042	0.640 ± 0.041	*	*
OS-TILs(mR+miR)	0.644 ± 0.062	0.671 ± 0.064	0.642 ± 0.035	0.668 ± 0.039	0.641 ± 0.050	0.677 ± 0.052	*	*
IMO-TILs	0.709 ± 0.030	0.737 ± 0.047	0.701 ± 0.034	0.760 ± 0.046	0.703 ± 0.029	0.746 ± 0.028	-	-

TABLE III

COMPARISON OF IMO-TILs WITH OTHER INTEGRATIVE SURVIVAL ANALYSIS MODELS. THE "*" IN EACH ROW INDICATES THAT THE PROPOSED IMO-TILs IS SIGNIFICANTLY SUPERIOR ($p < 0.05$) TO THE COMPETING METHOD VIA T-TEST

	BRCA		LUSC		KIRC		p-value	
	CI	AUC	CI	AUC	CI	AUC	CI	AUC
Cheng et al [15]	0.646 ± 0.057	0.667 ± 0.033	0.655 ± 0.028	0.671 ± 0.041	0.646 ± 0.036	0.669 ± 0.051	*	*
Gpdbn [40]	0.661 ± 0.038	0.687 ± 0.036	0.648 ± 0.051	0.677 ± 0.051	0.661 ± 0.043	0.679 ± 0.041	*	*
OSCCA [41]	0.659 ± 0.035	0.677 ± 0.041	0.651 ± 0.031	0.681 ± 0.052	0.660 ± 0.041	0.681 ± 0.039	*	*
IMGFN [21]	0.671 ± 0.031	0.694 ± 0.039	0.668 ± 0.021	0.704 ± 0.037	0.675 ± 0.031	0.704 ± 0.041	*	*
IMO-TILs-V1	0.694 ± 0.043	0.718 ± 0.059	0.682 ± 0.031	0.709 ± 0.035	0.679 ± 0.021	0.711 ± 0.034	*	*
IMO-TILs-V2	0.677 ± 0.044	0.704 ± 0.040	0.666 ± 0.029	0.693 ± 0.022	0.665 ± 0.027	0.699 ± 0.035	*	*
IMO-TILs	0.709 ± 0.030	0.737 ± 0.047	0.701 ± 0.034	0.760 ± 0.046	0.703 ± 0.029	0.746 ± 0.028	-	-

without incorporating the multi-omics data. 5) O-TILs(mR) or O-TILs(miR): A variant of the proposed IMO-TILs, which applies Concrete AutoEncoder on mRNA or miRNA data for cancer survival analysis. 6) OS-TILs(mR+miR): A variant of the proposed IMO-TILs by the combination of mRNA and miRNA data without incorporating image data. The symbol "*" shown in each row of Table II indicate that IMO-TILs is significantly superior ($p < 0.05$) to the corresponding competing method via t-test.

As can be seen from Table II, the combined measurements of pathological images and multi-omics data can consistently achieve more accurate prognosis results. Specifically, our integrative survival analysis model can achieve the CI values of 0.709, 0.701 and 0.703, and AUC values of 0.737, 0.760 and 0.746 on BRCA, LUSC and KIRC datasets, respectively, which are significantly higher than the approaches that either apply TILs spatial information derived from the pathological images or single-type of omics data and the combination of multiple omics data for clinical outcome prediction. Furthermore, we observe that OS-TILs can achieve higher prognosis results than the methods based on individual omics data (*i.e.*, O-TILs (mR), O-TILs (miR)). These results further demonstrates the effectiveness of combining multi-omics data for survival prediction. In addition, it is noteworthy that the proposed IMO-TILs and its variant I-TIL can achieve better prognosis results than the methods [6], [16] only considering the TILs distribution in pathological images for clinical outcome

prediction. The reason lies in the fact that our method also takes the intrinsic correlation between TILs and tumor within WSIs into consideration, which will benefit us in capturing the interactions between tumor and the host immune system for cancer survival analysis. From Table II, we also observe that the O-TILs method is consistently superior to the O-DeepSurv method on individual omics data. This is because the O-TILs model adopts the Concrete AutoEncoder framework to identify survival-associated features for the reconstruction of the input data, which is beneficial for the following survival analysis task.

C. Comparison of IMO-TILs With Other Integrative Survival Analysis Models

Next, in order to further demonstrate the effectiveness of our proposed method, we also compare IMO-TILs with the following five integrative survival analysis models in Table III. 1) Cheng et al. [15]: An integrative survival analysis method, which combines the cell-level images features and the mRNA data. 2) Gpdbn [40]: An attention based multi-modal fusion algorithm combining the pathological images and mRNA data for clinical outcome prediction. 3) OSCCA [41]: A canonical correlation analysis (*i.e.*, CCA) based multi-modal fusion framework combining the pathological images with mRNA data for cancer survival analysis. 4) IMGFN [21]: Our previous study published at MICCAI 2022, which integrate the

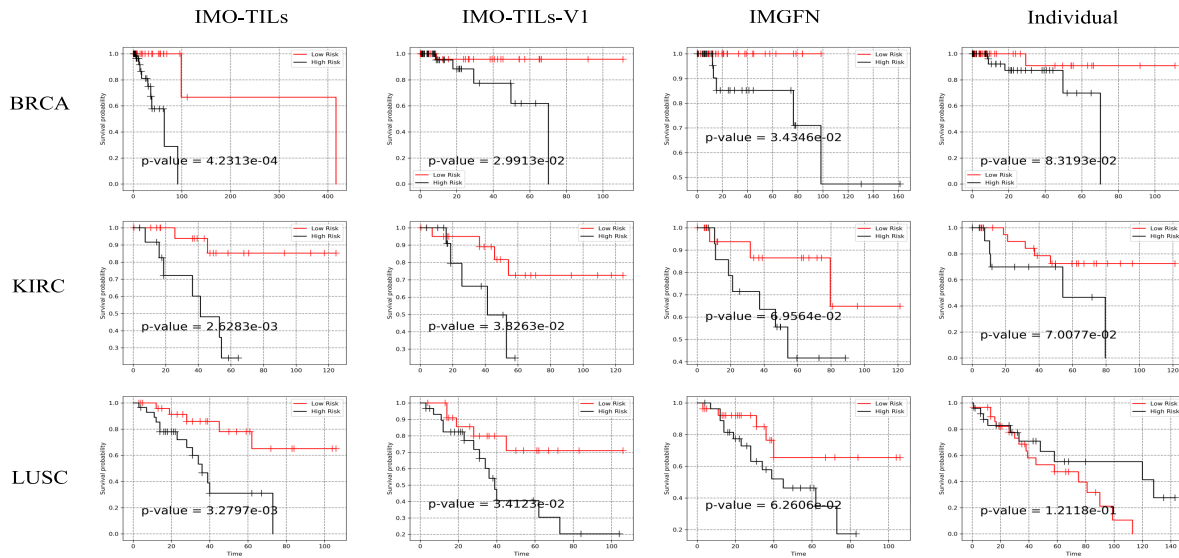


Fig. 2. Comparison of different methods for patients stratification.

spatial TIL-Tumor interactions with mRNA data for clinical outcome prediction. 5) IMO-TILs-V1: A variant of the proposed IMO-TILs, which directly combines the multi-modal feature together without applying the DGCCA to capture the association among them. 6) IMO-TILs-V2: A variant of the proposed IMO-TILs method combining the histopathological images with the miRNA data for survival prediction.

From Table III, we can observe that IMO-TILs and its variant IMO-TILs-V1 generally outperform their competitors only integrating pathological images with mRNA or miRNA data for human survival analysis. These results clearly demonstrate the advantage of the integrative analysis of image and multi-dimensional genomic data for patient outcome prediction. On the other hand, Table III also indicates that the approaches (*i.e.*, IMGFN, IMO-TILs-V1, IMO-TILs-V2 and IMO-TILs) that consider the spatial interactions between TILs and tumor patches can generally lead to the improvement on different cancer cohorts. These results again verify the fact that the investigation of TILs and their spatial correlation with tumors on whole-slide pathological images can better predict the patients' survival than the comparing methods [15], [40], [41] without considering it. Finally, as shown in Table III, the proposed IMO-TILs model can better predict the patients' survival than its variant IMO-TILs-1. This implies that capturing the associations among multi-modal data can help promote the final prognosis results.

D. Comparison of IMO-TILs With Other Methods for Patient Stratification

In the survival analysis of human cancers, another important task is the division of patient populations into different survival subgroups, which can help physicians design personalized therapeutic interventions on individuals. Specifically, we followed the method in [41] that applied the K-means clustering algorithm to aggregate the fused multi-modal features into 2 distinct sub-groups (*i.e.*, high and low survival risk groups). Then, the log-rank test [41] is applied to see if these two groups have significant difference of clinical outcome, and we only show the fold with the best stratification performance for different methods.

As can be seen from Fig. 2, the proposed IMO-TILs method can achieve significantly better patient stratification performance than the comparing methods. Here, the column of "Individual" indicates the method that can yield the best stratification performance for each type of data. Specifically, the p-values for our IMO-TILs methods on BRCA, KIRC and LUSC datasets are $4.23e-4$, $2.62e-3$ and $3.28e-3$, respectively. These results further demonstrate the advantage of the integrative analysis of image and multi-dimensional genomic data for patient outcome prediction. It is noteworthy that IMO-TILs can achieve better stratification performance than its variant IMO-TILs-V1, indicating that we can better predict the clinical outcome of cancer patients by further taking the correlation among multi-view data into consideration.

E. The Difference of the TILs-Tumors Intersections Between High-Survival and Low-Survival Risk Patients

In this section, we will discuss the difference of the TILs-Tumors intersections between the patients with high-survival and low-survival risks indicated by Fig. 2. Specifically, for the stratified patients in different groups, we check the important edges in each group according to their weights calculated via GAT. We select the top 100, 200, 300, 500 edges with the largest weights, and calculate the proportion of edges that connect TILs and tumor patches on BRCA, LUSC and KIRC datasets, respectively. As can be seen from Fig. 3, the proportion of edges that connect TILs and tumor regions in low-risk group are higher than that in high-risk group, showing that the interactions of TILs and tumors play a critical role in prognosis prediction of human cancers. Such observations are also consistent with the existing knowledge that the stronger interactions between tumor and TILs are a good prognostic factor for overall survival prediction of cancer patients, and vice versa [4].

In addition, we also visualize the sampled tumor and TILs patches and their connected edges in Fig. 4. Here, the thick black lines indicate the edges with higher weights calculated via GAT. As can be seen from Fig. 4, the TILs density in low-survival risk patients are higher than that in high-survival risk patients, indicating the TIL density is an important indicator of

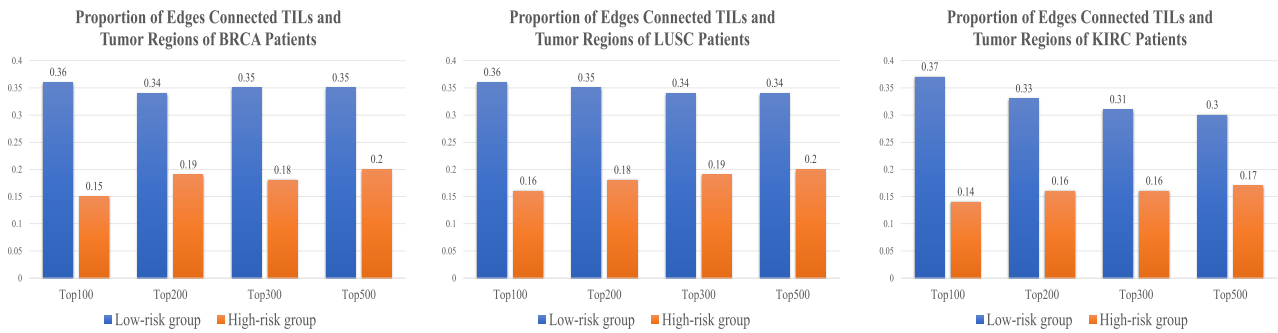


Fig. 3. Proportion of edges connected TILs and tumor patches on different cancer cohorts.

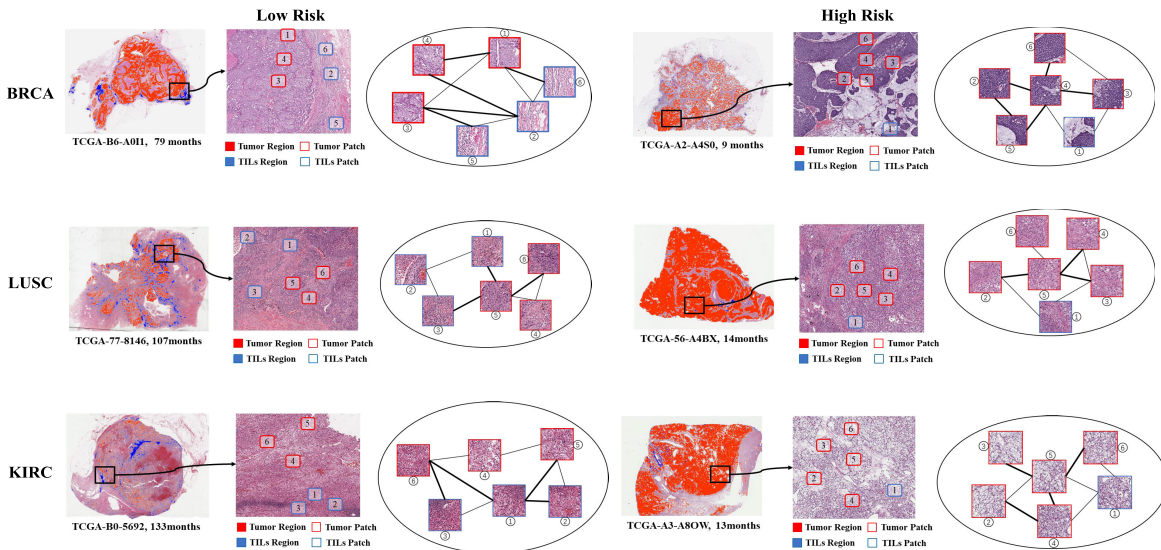


Fig. 4. The interactions between TILs patches and tumor patches for low survival risk and high survival risk patients. The thick black line indicates the edges with higher weights calculated via Graph Attention Network (GAT).

the patient outcomes. In addition, it is obvious that the weights of the edges that connect tumor and TIL regions are higher for patients in low survival risk group on all of the three cancer cohorts, which again validate the fact that the brisk interactions between TIL and tumor regions can regulate tumor growth by continuous immunosurveillance [43].

F. Identification of Important Multi-Omics Bio-Markers for Cancer Prognosis

Besides the better prognosis result, our method can also identify important multi-omics features via the utilized Concrete selector layers. For each type of the multi-omics data, we identify 10 survival-associate Eigengene features. Since the selected features in each fold are different, we report the selected features that are appeared in most folds of the cross-validation, and the results are shown in Table IV.

As can be seen from Table IV, three mRNA and three miRNA Eigengene features are identified for BRCA dataset. For mRNA features, the gene enrichment analysis by ToppGene (<https://toppgene.cchmc.org/>) indicates that Eigengene 20 is consisted of 27 genes, and 20 of them are associated with the regulation of the innate immune response on breast cancer ($p_value = 5.670e - 23$), and the recent genetic studies have resurrected the concept that the adaptive and innate immune systems play important roles in tumor growth [44]. In addition, Eigengene 40 is composed of 15 genes, and seven of them are correlated with tumor vasculature ($p_value = 1.374e - 5$). The recent study demonstrated that the structurally and

TABLE IV
IDENTIFICATION OF IMPORTANT MULTI-OMICS BIO-MARKERS FOR CANCER PROGNOSIS

DataSet	Feature Type	Selected Eigengene Features
BRCA	mRNA Feature	Eigengene 20, Eigengene 40, Eigengene 11
	miRNA Feature	Eigengene 7, Eigengene 10, Eigengene 11
KIRC	mRNA Feature	Eigengene 10, Eigengene 32, Eigengene 1, Eigengene 42
	miRNA Feature	Eigengene 2, Eigengene 17, Eigengene 1, Eigengene 6
LUSC	mRNA Feature	Eigengene 28, Eigengene 47
	miRNA Feature	Eigengene 1, Eigengene 3, Eigengene 4

functionally aberrant tumor vasculature can simultaneously maintain cancer cells' permissive environment and generating a physical barrier to T cells' infiltration [45]. For miRNA feature, Eigengene 10 is consisted of 5 genes, and three of them (*i.e.*, hsa-mir-140, hsa-mir-204, hsa-mir-195) are correlated with regulation of tumor necrosis factor (TNF) production ($p_value = 1.416e - 5$). In breast cancer, TNF correlates with augmented tumor cell proliferation, which increases the occurrence of metastasis and are generally recognized as poor prognosis indicators [46]. Moreover, Eigengene 11 contains four genes, and two of them (*i.e.*, hsa-mir-20a and hsa-mir-17)

TABLE V
THE EFFECTS OF PARAMETER λ FOR PROGNOSIS PREDICTION

Triplets Number	$\lambda = 1e-2$	$\lambda = 1e-3$	$\lambda = 1e-4$	$\lambda = 1e-5$
BRCA	0.655	0.709	0.695	0.687
KIRC	0.652	0.701	0.689	0.681
LUSC	0.650	0.703	0.678	0.687

are involved in B cell apoptotic process, which is essential for the development and homeostasis of the human immune system [47]. Finally, Eigengene 7 is consisted of five genes, and two of them (*i.e.*, hsa-mir-128-1 and hsa-mir-128-2) are associated with the promotion of anti-tumor immunity in breast cancer (p -value= 9.983e-9) [48].

As to KIRC cohort, four Eigengenes are identified for the mRNA data. Specifically, Eigengene 10 is consisted of 58 genes, and 15 of them are correlated with the development of KIRC (*i.e.*, $p_value = 4.012e-4$). Among these 15 genes, the over-expression of MAL2 correlates with immune infiltration, which often yields with better prognosis results [49], while the patients with lower expression of EpCam showed a trend toward worse prognosis for KIRC patients. In addition, seven genes in Eigengene 32 correlates with pyrophosphatase (PPA2) activity, and the down-regulation of PPA2 expression correlates with poor prognosis of KIRC [50]. Moreover, the identified Eigengene 42 is consisted of four genes (*i.e.*, SLC39A6, FASN, NPY1R, GATA3), and it is widely recognized that GATA3 is highly correlated with the luminal transcriptional program in KIRC [51]. For miRNA data, all of the three genes in Eigengene 17 correlates with the negative regulation of lymphocyte apoptotic process, and four genes in Eigengene 6 is associated with the biological process of epithelial cancer cell migration, and the invasion of cancer cells into surrounding tissue and the vasculature is an initial step in tumor metastasis [52].

Finally, for the mRNA data in LUSC cohort, Eigengene 28 is consisted of 8 genes, and six of them (TARS1, BRX1, CCT5, NSUN2, PAIP1, NUP155) are correlated with the process of UFMylation, which is widely accepted as a crucial post-translational modification for maintenance of p53 stability and tumour-suppressive function [53]. In addition, the existing study has demonstrated that the expression of ERLIN2 in Eigengene 47 is closely associated with immune infiltrates of TILs [54]. As to miRNA features, four genes (*i.e.*, hsa-mir-16-1, hsa-mir-15a, hsa-mir-497, hsa-mir-195) in Eigengene 1 correlates with the immune cell development and homeostasis. Moreover, the existing studies have demonstrated that three genes (hsa-mir-509-1 hsa-mir-509-3 hsa-mir-509-2) in Eigengene 4 can suppress the proliferation, migration, and invasion of LUSC.

In summary, the identified multi-omics features on all of the three cancer cohorts are associated with the immune mechanism of TILs as well as their intersections with Tumors, which are consistent with the image analysis results described in Section III-F.

G. Influence of Parameter

In the proposed IMO-TILs method, only one parameter λ (shown in Eq.(16)) need to be tuned. In this section, we discuss

TABLE VI
COMPARISONS OF THE PROPOSED IMO-TILs WITH ITS COMPETITORS TRAINED BY THE COMBINATION OF IMAGE AND miRNA DATA

	BRCA	LUSC	KIRC
Cheng [15]	0.635 $\pm$ 0.029	0.622 $\pm$ 0.021	0.639 $\pm$ 0.032
Gpdbn [40]	0.654 $\pm$ 0.031	0.655 $\pm$ 0.047	0.667 $\pm$ 0.051
OSCCA [41]	0.648 $\pm$ 0.021	0.654 $\pm$ 0.036	0.653 $\pm$ 0.044
IMGF [21]	0.662 $\pm$ 0.065	0.659 $\pm$ 0.044	0.671 $\pm$ 0.039
IMO-TILs	0.709 $\pm$ 0.030	0.701 $\pm$ 0.034	0.703 $\pm$ 0.029

how IMO-TILs performs with different values of λ on BRCA, LUSC and KIRC datasets, and record their CI values in Table V. From Table V, we can clearly see that the CI values are affected with different λ values on all of the three cancer cohorts, which implies that the selection of λ is very important for the final prognosis results. This is reasonable since λ controls the weights for capturing the association among multi-modal data, and the small values of λ (*i.e.*, $\lambda = 1e-4$, $1e-5$) yield to inferior prognosis results since they cannot make full use of such association for multi-modal data integration. On the other hand, the CI values are significantly decreased if λ is increased to $1e-2$. This is because the larger values of λ will weaken effects of the survival loss (*i.e.*, first term in Eq. (16)), and thus leading to inferior prognosis results.

IV. DISCUSSION

A. Comparisons of the Proposed IMO-TILs With Its Competitors Trained by the Combination of the Histopathological Image and miRNA Data

As discussed in Section III-C, most of the integrative analysis models (*i.e.*, [15], [21], [40], and [41]) are based on the combination of histopathological image and mRNA data for survival prediction. In this section, we further compare their predictive power with IMO-TILs if they are trained with the histopathological image and miRNA data, and the Concordance Index (CI) of different methods are shown in Table VI.

As can be seen from Table VI, the proposed IMO-TILs can still achieve better prognosis results when comparing with the variants of the existing studies that combine the histopathological images with the miRNA data for patients' clinical outcome prediction. These results again validate the fact that miRNA and mRNA data complement with each other, and thus the prognosis results will be further improved if we combine them together with the image data for the survival prediction task.

TABLE VII

DEMOGRAPHICS AND CLINICAL INFORMATION OF THE DATASET				
	TNBC	Her2+	Luminal A	Luminal B
Number of Patients:	41	40	139	199
Concordance Index	0.667	0.636	0.705	0.708

B. Computational Complexity Analysis of the Proposed IMO-TILs Method

In this section, we discuss the computational complexity of the proposed IMO-TILs method. Actually, the time complexity of IMO-TILs is dominated by three components, *i.e.*, the Graph Attention Networks (GATs) implemented on the image data, the Concrete Auto-Encoder worked on each individual genomic data (*i.e.*, mRNA and miRNA), and the application of the Deep Generalized Canonical Correlation Analysis (DGCCA) for the integrative survival analysis of human cancers. Specifically, the time complexity of the GAT attention layer can be expressed as $O(N(|V|d_I o_I + |E|o_I))$, where N is the number of samples, $|V|$ and $|E|$ denote the number of edges and nodes in each graph, while d_I and o_I correspond to the dimensionality for the input and output of the GAT layer. As to the concrete autoencoder layer applied on mRNA or miRNA data, the time complexity is denoted as $O(Nd_m t k)$ or $O(Nd_{mi} t k)$, where d_m and d_{mi} refer to the dimensionalities of the input mRNA and miRNA features, respectively, t represents the dimension of the concrete random variable used for identifying survival-associated genomic features, and k defines the number of features to be selected. Finally, the gradient descent update for the DGCCA model has the time complexity $O(NJrd_o)$, where r is the dimension for the shared representation in the DGCCA model, J refers to the number of views, and d_o is the largest dimensionality of the input views. Suppose d_{max} to be the largest dimensionality among input and output data for all the views. Hence, the time complexity of the proposed IMO-TIL model is $O(Nd_{max}(|V|d_{max} + |E| + Jr + 2tk))$.

C. The Prognosis Performance of IMO-TILs on Different Sub-Types of Breast Cancer

In this section, we discuss the prognosis performance of the proposed IMO-TILs on different subtypes of breast cancers (*i.e.*, TNBC, HER2+, Luminal A and Luminal B). The results are shown in Table VII of the revised manuscript. For your convenience, we also list it as follows:

As shown in Table VII, on one hand, the proposed IMO-TIL method can still achieve satisfied Concordance Index (CI) of 0.705 and 0.708 for Luminal A and Luminal B patients, respectively, which are also similar to the prognosis results of IMO-TILs applied on the whole BRCA cohort (shown in Table III). These results further verified the advantage of our method that combines the spatial interactions between tumor and lymphocytes with multi-omics data for prognosis predictions of human cancers. On the other hand, we noticed that the prognosis result on TNBC and HER2+ cohorts are inferior to that on Luminal A and Luminal B patients. One possible reason is that the number of TNBC and HER2+ samples in the BRCA cohort are significantly smaller than the other two breast cancer subtypes, which will affect the evaluation of the proposed prognosis model.

V. CONCLUSION

In this paper, we develop IMO-TILs, an multi-modal fusion framework that can combine pathological images with multi-omics data for prognosis predictions of human cancers. The main contribution of our approach is fusing the interaction information between TILs and tumors with the multi-omics data via an attention mechanism, which is proved to be more accurate in prognosis predictions as well as better stratification of patients than comparing methods. In addition, we characterize the spatial interactions of TILs and tumor regions across WSIs by GAT and find that patients with long survival time are demonstrated to have stronger association between TILs and tumor regions than short survival patients. Meanwhile, the multi-omics feature selected by Concrete Auto-encoders are proved to be associated with TILs and tumors as well as their interactions for cancer survival analysis. In summary, we propose an easy-to-implement general framework that can characterize the survival-associated interactions between tumor-infiltrating lymphocytes (TILs) and tumors from pathological images and multi-omics data with good explainability. It is worth noting that though the simple implementation of the proposed approach achieves good results to demonstrate our idea, we can design more advanced techniques to replace the existing modules (*e.g.*, CAE and DGCCA) for further performance improvement.

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